

$$A=\mu(O)\mu$$

$$\mu_1\mu_5$$

$$\times$$

$$A = \mu(O)$$

$$A = \mu(O)$$

$$\mu_1\mu_5$$

$$\mu_1$$

μ_2

μ_3

μ_4

μ_5

$$\mu_1\mu_5$$

μ_1

$$S(t)tI(t)t\phi(t)$$

$$\phi(t)=\frac{|S(t)\cap I(t)|}{|S(t)\cup I(t)|}$$

$$|\cdot|$$

$$\phi(t)=\phi_0e^{-\lambda t}$$

$$\phi_0\approx 0.95\lambda\approx 0.15$$

$$\phi(12)=0.95e^{-0.15\times 12}\approx 0.95\times 0.165\approx 0.157$$

$\neq$

$$2^{32} \approx 4.3 \times 10^9$$

$$nD_i$$

$$N_{tests} = \prod_{i=1}^n |D_i|$$

$$\frac{N_{tests}}{\prod |D_i|} \ll 1$$



$$A=\mu(O)$$

$$OA\mu:O\rightarrow AOA$$

$$A=\mu(O)$$

$$\mu\mu$$

$$\begin{array}{c}\mu\\ \mu\end{array}$$

$$\mu=\mu_5\circ\mu_4\circ\mu_3\circ\mu_2\circ\mu_1$$

$$\mu_1$$

$$\mu_2$$

$$\mu_3$$

$$\mu_4$$

$$\mu_5$$

$$O\mu(O)A\mu OA$$

$$\mu\mu=\mu_5\circ\mu_4\circ\mu_3\circ\mu_2\circ\mu_1\mu_i$$

$$\begin{array}{l} A_1=\mu(O)\\ A_2=\mu(O)\\ \implies A_1=A_2 \end{array}$$

$$A=\mu(O)OAA\mu(O)$$

$$R(h,\sigma,t,m)$$

$$h=H(A)A$$

$$\sigma = {}_{sk}(h)$$

$$t$$

$$m$$

$$RAO$$

$$h' = H(A')A'$$

$$h'h$$

$$\sigma$$

$$AO$$

$$R(h,\sigma,t)$$

$$h = H(A)A$$

$$\sigma = {}_{sk}(h)$$

$$t$$

$$RAO$$

$$h' = H(A')A'$$

$$h'h$$

$$\sigma$$

$$AO$$

×

$$n \geq 1000$$

✓

✓

✓

✓

✓

✓

✓

✓

✓

✓

✓

✓

✓

✓

✓

✓

$\times \times$

$$A = \mu(O)$$

$\times$

$$\begin{array}{l} O \\ A \\ \mu \\ H(O) \leq 20 \\ \Phi(O,A) = I(O;A)/H(O) \end{array}$$

$$O,O_1,O_2,\ldots$$

$$A,A_1,A_2,\ldots$$

$$\mu,\mu_i,f,g,\ldots$$

$$\mathcal{O},\mathcal{A},\mathcal{T},\ldots$$

$$H(O)$$

$$I(O;A)$$

$$H(A),h_1,h_2,\ldots$$

$$\sigma,\sigma_1,\sigma_2,\ldots$$

$$A=\mu(O)$$

$$\times$$

$$S=(O,t,V,G)$$

$$O\in\mathcal{O}$$

$$t\in\mathbb{T}$$

$$V\in\mathcal{V}$$

$$G\in\mathcal{G}$$

$$\mathcal{O}$$

$$O_t=\{a_1,a_2,\ldots,a_n\}$$

$$a_i$$

$$(a_i)=H(a_i)$$

$$\mathbb{T}$$

$$t(e) = (t(e_{prev}) + 1, t(remote) + 1)$$

$$\begin{array}{c} e_1e_2e_1 \rightarrow e_2t(e_1) < t(e_2) \\ \mathcal{V} \end{array}$$

$$V(e)=\langle v_1,v_2,\ldots,v_n\rangle$$

$$v_i i$$

$$e_1\parallel e_2\iff V(e_1)\not\leq V(e_2)\wedge V(e_2)\not\leq V(e_1)$$

$$\mathcal{G}$$

$$G=\{(commit,hash),(branch,commit),(tag,commit)\}$$

$$S=(O,t,V,G)S'$$

$$S'=(S,\Delta t,\Delta V,\Delta G)$$

$$\Delta$$

$$t'O_{t'}$$

$$O_{t'} = \{a \in O_t \mid (a) \leq t' \wedge \neg(a) \leq t'\}$$

$$(t')=(events,t')$$

$$(V)=(dependencygraph(V))$$

$$(G,t')=(H(commit_{t'}))$$

$$St$$

$$\forall m_1,m_2:(S,m_1)=(S,m_2)$$

$$So_1,o_2t$$

$$O_{o_1,t}=O_{o_2,t}$$

$$\mu:O\rightarrow AOA=\mu(O)$$

$$\mu\mu(O)OOO_{o_1,t}=O_{o_2,t}$$

$$\begin{aligned} A_{o_1,t} &= \mu(O_{o_1,t}) \\ &= \mu(O_{o_2,t})O \\ &= A_{o_2,t} \end{aligned}$$

$$A$$

$$A\overset{OA}{=} \mu(O)$$

$$\mu:\mathcal{O}\rightarrow\mathcal{A}$$

$$\mu=\mu_5\circ\mu_4\circ\mu_3\circ\mu_2\circ\mu_1$$

$$\mu_1$$

$$\mu_2$$

$$\mu_3$$

$$\mu_4$$

$$\mu_5$$

$$\mu$$

$$O\mu(O)A$$

$$\begin{array}{c} \mu_i \\ f:X\rightarrow Y \end{array}$$

$$y=f(x)$$

$$f$$

$$f$$

$$\mu_1$$

$$A_1=\mu_1(O)=((O))$$

$$\mu_1$$

$$\mu_2$$

$$A_2=\mu_2(A_1)=(A_1,)$$

$$\mu_2$$

$$\mu_3$$

$$A_3=\mu_3(A_2)=(A_2,)$$

$$\mu_3$$

$$\mu_4$$

$$A_4=\mu_4(A_3)=((A_3))$$

$$\mu_4$$

$$\mu_5$$

$$A_5=\mu_5(A_4)=((A_4))$$

$$\mu_5$$

$$\mu_i$$

$$A_1=\mu_1(O)$$

$$A_2=\mu_2(A_1)$$

$$A_3=\mu_3(A_2)$$

$$A_4=\mu_4(A_3)$$

$$A=\mu_5(A_4)=\mu(O)$$

$$\mu(O)$$

$$Oe_1,e_2$$

$$\mu_{e_1}(O)=\mu_{e_2}(O)$$

$$O$$

$$H(O)=-\sum_i p_i\,{}_2p_i$$

$$p_i\dot{t}$$

$$H(O)\approx 0H(O)\gg 0$$

$$H(O)\leq 20\mu(O)$$

$$K(O)$$

$$K(O)\leq H(O)+O(1)$$

$$H(O)\leq 202^{20}\approx 1$$

$$H(O)\leq 20$$

$$H(O)>20$$

$$O$$

$$H(O)\leq 20$$

$$\forall e\in \mathcal{O}:(e)\subseteq O$$

$$\forall a\in \mu(O):a\models \Sigma$$

$$H(O)=-\sum_i p_i\,2^{-p_i}\leq 20$$

$$\forall e\in \mathcal{O}:(e)\subseteq O$$

$$\forall a\in \mu(O):a\models \Sigma$$

$$\Sigma$$

$$O(|O|\cdot |(O)|)$$

$$O(|O|)$$

$$O(|\cdot|\cdot|O|)$$

$$O(|\mu(O)|)$$

$$||\approx |(O)|O(|O|\cdot |(O)|)$$

$O\Sigma$

$H \leftarrow (O)$
 $H > 20$

▷

$e \in$
 $\leftarrow (O, e)$
 $=$

▷

$A \leftarrow \mu(O)$
 $a \in A$
 $\neg(a \models \Sigma)$

▷

$$h_cO$$

$$h_c=H(O)\oplus H((O))\oplus H((O))$$

$$H\oplus$$

$$O_1,O_2\mu(O_1)=\mu(O_2)$$

$$\begin{array}{l} H(O_1)=H(O_2)\\ H((O_1))=H((O_2))\\ H((O_1))=H((O_2)) \end{array}$$

$$O_1=O_2\mu(O_1)=\mu(O_2)$$

$$\rightarrow$$

$$OTE A = \mu(O)TE$$

$$_O(E)=T\implies _A(E)=T$$

$$\mu_1$$

$$\mu_2$$

$$\mu_3$$

$$\mu_4$$

$$\mu_5$$

$$\mu=\mu_5\circ\mu_4\circ\mu_3\circ\mu_2\circ\mu_1A(E)=_O(E)$$

$$\forall a\in A=\mu(O):(a)=_O(a)$$

$$OA$$

$$\Phi(O,A)=\frac{I(O,A)}{H(O)}$$

$$I(O,A)OAH(O)$$

$$\Phi(O,A)=1.0$$

$$\Phi(O,A)=0.0$$

$$\Phi(O,A)=0.98$$

$$I(O,A)=H(O)-H(O|A)$$

$$H(O|A)OAH(O|A)=0I(O,A)=H(O)\Phi(O,A)=1.0$$

$$\Phi(O,A)\geq 0.98$$

$$\rightarrow$$

$$\leq 1\Phi(O,A)\geq 0.99$$

$$\Phi(O,A)<0.98$$

$$O$$

$$O$$

$$R$$

$$R=(h,\sigma,\tau,m)$$

$$h=H(A)$$

$$\sigma = {}_{sk}(h)$$

$$\tau$$

$$m$$

$$C = [R_1, R_2, \ldots, R_n]R_iR_{i-1} \qquad R_i.h_{prev} = H(R_{i-1})$$

$$A_1R_1$$

$$A_2R_2R_2.h_{prev} = H(R_1)$$

$$R_n$$

$$C$$

$$R_i.h_{prev} = H(R_{i-1})R_{i-1}H(R_{i-1})$$

$$\forall i \in [1, n-1] : (R_i.h_{prev} = H(R_{i-1}))$$

$$O(n)$$

$$T$$

$$h_i = H(R_i)$$

$$h_{i,j}=H(h_i||h_j)$$

$$h_{root}$$

$$R_i$$

$$h_i = H(R_i)$$

$$O(n)$$

$$RskAO$$

$$\sigma = {}_{sk}(h)skpk$$

$$sk$$

$$h$$

$$\tau$$

$$AO\tau$$

$$A=\mu(O)$$

$ARpk$

$h' \leftarrow H(A)$
 $h' \neq R.h$ ▷

$\leftarrow_{pk}(R.h, R.\sigma)$
 $\neg$ ▷

$\leftarrow ()$
 $R.\tau >$ ▷

$\neg(R.m.)$ ▷

f

$O(n^2)$

$\leftrightarrow$

$\rightarrow$

$$O(n^2)$$

$$1 + n^2 + n^2 = O(n^2)$$

$$O(n)$$

→

$$\mu_1\mu_5$$

$$\mu=\mu_5\circ\mu_4\circ\mu_3\circ\mu_2\circ\mu_1$$

$$\mu_1$$

$$\mu_2$$

$$\mu_3$$

$$\mu_4$$

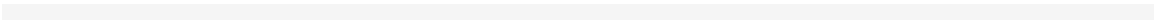
$$\mu_5$$

$$\mu=\mu_5\circ\mu_4\circ\mu_3\circ\mu_2\circ\mu_1\mu_i$$

$$\mu_1\mu_1(O)\mu_2\mu_2(\mu_1(O))$$



$$\mu_1$$



μ_1

input

graph $\leftarrow$ (*input*)

errors $\leftarrow$ (*graph*)

errors $\neq \emptyset$

(*errors*)

entropy $\leftarrow$ (*graph*)

entropy > 20

()

(*graph*, *entropy*)

C(*target*, *condition*, *severity*)

target

condition

severity

$\mu_1(O)O$

graph

reference $\in$ (*graph*)

definition $\leftarrow$ (*graph*, *reference*)

definition =

(+ *reference*)

()

$$H(G)=-\sum_{i=1}^n p(x_i)_2p(x_i)$$

$$p(x_i)x_i$$

$$C(target,condition,severity)$$

$$target$$

$$condition$$

$$severity$$

$$\mu_1(O)O$$

<i>graph</i>
<i>reference</i> $\in$ (<i>graph</i>)
<i>definition</i> $\leftarrow$ (<i>graph</i> , <i>reference</i>)
<i>definition</i> =
(+ <i>reference</i>)
()

 $\mu_1\mu_5$

μ_2

Q

$$Q : G_{input} \rightarrow G_{output}$$

$QGQ(G)$

property, graph

ranges $\leftarrow$ (*property, graph*)

usages $\leftarrow$ (*property, graph*)

$|ranges| = 1$

ranges[0] $\triangleright$

$|usages| > 0$

inferred $\leftarrow$ (*usages*) $\triangleright$

inferred $\triangleright$

$T : Data \rightarrow String$

extracted, prefixes

ctx $\leftarrow ()$

ctx.insert(entities, extracted.entities)

ctx.insert(relationships, extracted.relationships)

ctx.insert(constraints, extracted.constraints)

ctx.insert(prefixes, prefixes)

ctx.insert(timestamp,)

ctx

$$\mu_1\mu_5$$

$$$$

$$$$

$$\iota type(rdf_{\iota type}) : ConvertRDFtypetoRusttype$$

$$$$

$$$$

$$\rule{1cm}{0.4pt}$$

$$\rule{1cm}{0.4pt}$$

$$\rule{1cm}{0.4pt}$$

$$\mu_4$$

$$C(A)A$$

$$AC(C(A)) = C(A)$$

$$C(C(A)) = C(A)$$

artifact, expectedhash
canonical $\leftarrow$ (*artifact*)
actualhash $\leftarrow$ (*canonical*)
actualhash = *expectedhash*
()

()

—
—

—
—
—
—

artifact, spechash
artifacthash $\leftarrow$ (*artifact*)
combined $\leftarrow$ *spechash*||*artifacthash*
signature $\leftarrow$ (*sk, combined*)
timestamp $\leftarrow$ ()
metadata $\leftarrow$ {
 version : 1.0,
 generator : *ggen*,
 pipelineversion : 6.0.0

receipt $\leftarrow$ (*artifacthash, signature, timestamp, metadata*)
receipt

$$R_{i.parent} = H(R_{i-1})$$

$$R_i R_j : j > i$$

$$R_j j > i H(R_{j-1}) R_i H(R_i) H(R_{i+1})$$

$$()$$

!valid

$$()$$
$$()$$

μ_1	$O(O)$	$O(O)$
μ_2	$O(O \cdot Q)$	$O(O + Q)$
μ_3	$O(E \cdot T)$	$O(E \cdot T)$
μ_4	$O(A \cdot A)$	$O(A)$
μ_5	$O(A)$	$O(1)$
$O(O \cdot Q + E \cdot T + A \cdot A)$		$O(O + E + A)$

$|O|$

$|Q|$

$|E|$

$|T|$

$|A|$

$O(|O| \cdot |Q|)$

$O(|E| \cdot |T|)$

$O(|A| \cdot |A|)$

$$|O|$$

$$|Q|$$

$$|E|$$

$$|T|$$

$$|A|$$

$$\rule{1.5cm}{0.4pt}$$

$$\rule{1.5cm}{0.4pt}$$

$$\rule{1.5cm}{0.4pt}$$

$$\rule{2.5cm}{0.4pt}$$

$$\rule{2.5cm}{0.4pt}$$

$$\rule{2.5cm}{0.4pt}$$

$$Ss$$

$$S=\frac{1}{s+\frac{1-s}{n}}$$

$$n$$

$$s=0.15$$

$$S=\frac{1}{0.15+\frac{0.85}{8}}=4.8\times$$

$$4.2\times$$

10 – 50×



spec, iterations
hashes $\leftarrow \emptyset$
i = *iterations*
artifact $\leftarrow \mu(\textit{spec})$
hash $\leftarrow (\textit{artifact})$
hashes.insert(hash)

$|\textit{hashes}| = 1$
()

(+ $|\textit{hashes}|$ +)

$$P() \approx 1 - e^{-\frac{n^2}{2 \cdot 2^{256}}}$$

$$n = 10,000$$

$$P() \approx 1 - e^{-\frac{10000^2}{2 \cdot 2^{256}}} \approx 0$$

$$SC\mu\mu_C(S) = \mu'_C(S)C = C'$$

$$\mu_1$$

$$\mu_2$$

$$\mu_1\mu_5$$

$$\mu_3$$

$$\mu_4$$

$$\mu_5$$

$$() = \implies () =$$



```
task
tool ← (task.toolname)
tool =
()

validated ← (task.args, tool.inputschema)
validated.iserr
()

result ← (tool, task.args)
result
```

```
openapi
tools ← ∅
path, methods ∈ openapi.paths
method, spec ∈ methods
tool ← {
  name : spec.operationId,
  description : spec.description,
  inputschema : spec.requestBody.schema,
  outputschema : spec.responses['200'].schema
```

```
tools.push(tool)
```

```
tools
```

task, tools

prompt $\leftarrow$ (*task, tools*)

response $\leftarrow$ (*prompt*)

parsed $\leftarrow$ (*response*)

parsed.iserr

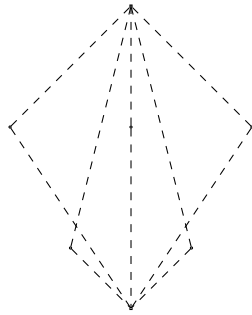
()

validated $\leftarrow$ (*parsed, task.outputschema*)

validated.iserr

()

parsed



```
task, timeout  
start ← ()  
result ← (task)  
!result.ready  
() − start > timeout  
()
```

```
(10ms)
```

```
result.get()
```

```
operation, maxretries  
retry = 0  
maxretries  
result ← operation()  
result.isok  
result  
  
delay ← 2retry × 100ms  
(delay)  
  
()
```

$$2f$$

$$2f+1$$

$$fn=3f+1$$

$$ff$$

$$n-f-f\geq f+1\implies n\geq 3f+1$$

$$f+1$$

```
artifact, spec
localvalid  $\leftarrow$  (artifact, spec)
vote  $\leftarrow$  (nodeid, localvalid, artifacthash)
(vote)
votes  $\leftarrow$  ( $2f + 1$ )
votes.filter(v => v.valid).length  $\geq 2f + 1$ 
()

()
```

$$O(n^2)$$

```
task
plan ← (task)
step ∈ plan
tool ← step.tool
args ← step.arguments
validation ← (tool, args)
validation.iserr
()

result ← (tool, args)

()
```

—
—

—
—

—

—
—
—

spec
referencehash $\leftarrow \mu(\textit{spec}).\textit{hash}()$
i = 110000
artifact $\leftarrow \mu(\textit{spec})$
hash $\leftarrow \textit{artifact}.\textit{hash}()$
hash $\neq \textit{referencehash}$
(+ *i*)

()

Λ

$$\Lambda(O) = (CPU_{limit}, Memory_{limit}, IO_{limit})$$

spec, metrics

limits $\leftarrow \Lambda(spec)$

metrics.cpuusage > 90

limits.cpu $\leftarrow limits.cpu \times 1.5$

metrics.memoryusage > 90

limits.memory $\leftarrow limits.memory \times 2$

limits

tasks

slots $\leftarrow (timewindow)$

task $\in tasks$

slot $\leftarrow (slots)$

slot.add(task)

slots

—

—

—

$$T_1T_2$$

$$(T_1) = (T_2)$$

system

metrics $\leftarrow$ (*system*)
opportunities $\leftarrow$ (*metrics*)
opportunities.isempty
(1*hour*)

opp $\in$ *opportunities*
optimized $\leftarrow$ (*opp*)
regenerated $\leftarrow$ μ (*optimized*)
(*regenerated*)

$G = (V, E)$

V

$E \subseteq V \times V(v_i, v_j)v_iv_j$

O_{new}, O_{old}, G

$\Delta \leftarrow (O_{new}, O_{old})$

▷

affected $\leftarrow$ (Δ, G)

▷

unchanged $\leftarrow V \setminus affected$

$A_{new} \leftarrow \mu(affected)$

▷

$A_{final} \leftarrow A_{new} \cup \mu(unchanged)$

▷

A_{final}

$$O(|V|)$$

$$O(|\Delta|\cdot d)d$$

$$20\times$$

$$\mu_1,\mu_2,\mu_3,\mu_4,\mu_5E_1,E_2,\ldots,E_n$$

$$\bigoplus_{i=1}^n \mu(E_i) = \mu\left(\bigcup_{i=1}^n E_i\right)$$

$$\oplus$$

$$E_i\mu$$

<i>spec</i>	
<i>parts</i> $\leftarrow$ (<i>spec</i> , <i>N</i>)	▷
<i>pipeline</i> $\leftarrow$ (<i>N</i>)	
<i>part</i> $\in$ <i>parts</i>	
<i>pipeline.submit</i> (<i>part</i> , <i>worker_i</i>)	
 <i>results</i> $\leftarrow$ <i>pipeline.collectall</i> ()	
<i>merged</i> $\leftarrow$ (<i>results</i>)	▷
<i>receipt</i> $\leftarrow$ (<i>merged</i>)	
<i>merged</i> , <i>receipt</i>	

$$3f+1f$$

$$k$$

$$(triple) = (subject) \enspace k$$

$$\begin{array}{l} O_{new}, O_{old} \\ \Delta \leftarrow (O_{new}, O_{old}) \\ view \in MaterializedViews \\ view.intersects(\Delta) \\ view.incrementalupdate(\Delta) \end{array}$$

$$O(n)O(n)$$

$$O(n^3)$$

$$C_{old}O_{old}\Delta$$

$$C_{new} = C_{old} \cup (\Delta \cup (C_{old}, \Delta))$$

$$(S)S$$

$$OA = \mu(O)$$
$$(O, A) = \{(o, a, h) \mid o \in O, a \in A, h = (a)\}$$

O, A	
$H_O \leftarrow (O)$	▷
$H_A \leftarrow (A)$	▷
$sig \leftarrow_{sk}(H_O \ H_A)$	▷
$receipt \leftarrow \{H_O, H_A, sig\}$	
$receipt$	

$receipt, O, A$
 $H'_O \leftarrow (O)$
 $H'_A \leftarrow (A)$
 $pk(receipt.sig, H'_O \| H'_A)$

$$OW = \mu_{WASM}(O)$$

$O, Telemetry$
 $usagepatterns \leftarrow (Telemetry)$
 $unusedentities \leftarrow \{e \in O \mid e.usagecount = 0\}$
 $hotpaths \leftarrow (usagepatterns)$
 $optimizations \leftarrow (hotpaths)$
 $O' \leftarrow O \setminus unusedentities \cup optimizations$
 O'

```

metrics  $\leftarrow ()$ 
metrics.latency  $> SLO$ 
opportunities  $\leftarrow (metrics)$ 
opp  $\in opportunities$ 
Ooptimized  $\leftarrow (opp)$ 
Anew  $\leftarrow \mu(O_{optimized})$ 
(Anew)

```

```

(monitorinterval)

```

$O\Sigma$

```

A =  $\mu(O)$ 
P = (A,  $\Sigma$ )
P  $\vdash A \models \Sigma$ 

```

```

 $\Box(request \rightarrow \Diamond response)$ 
 $\Box(state = Ready \rightarrow \bigcirc state \neq Busy)$ 
 $\Diamond(state = Done)$ 

```

O, LTL	
$M \leftarrow (O)$	$\triangleright$
$properties \leftarrow LTL$	$\triangleright$
$prop \in properties$	
$result \leftarrow (M, prop)$	
$result = Violated$	
$counterexample \leftarrow (M, prop)$	
$Err(counterexample)$	

Ok

$$T_1, T_2, \ldots, T_n T = T_1 \circ T_2 \circ \cdots \circ T_n$$

$$T(data) = T_n(T_{n-1}(\cdots T_2(T_1(data)) \cdots))$$

$$T_1$$

$$T_2$$

$$T_3$$

$$T_4$$

$$T_5$$

$$T_{base}DT_{spec}T_{base}S \subseteq DT_{spec}S$$

$$receipt.sig = Dilithium(H_O || H_A)$$

$$O$$

$$algorithm \leftarrow (O)$$

$$qasm \leftarrow (algorithm)$$

$$\triangleright$$

$$circuit \leftarrow (qasm)$$

$$simulator \leftarrow (circuit.qubits)$$

$$result \leftarrow simulator.run(circuit)$$

$$qasm, result$$

M, Φ
 $\phi \in \Phi$
 $\phi.type = Robustness$
 $verified \leftarrow (M, \phi.bound)$
 $\phi.type = Fairness$
 $verified \leftarrow (M, \phi.metric)$
 $\phi.type = Monotonicity$
 $verified \leftarrow (M, \phi.inputs)$

$\neg verified$
 $Err(\phi, counterexample)$

Ok

$M, \mathcal{H}, Objective$	
$samples \leftarrow (\mathcal{H}, k = 10)$	▷
$h \in samples$	
$score \leftarrow Objective(M(h))$	▷
$\mathcal{D}.add((h, score))$	▷
$surrogate \leftarrow (\mathcal{D})$	
$h_{next} \leftarrow (surrogate, \mathcal{D})$	
$score \leftarrow Objective(M(h_{next}))$	
$\mathcal{D}.add((h_{next}, score))$	
$h^* \leftarrow \mathop{\mathrm{argmax}}_{(h,s) \in \mathcal{D}} s$	
h^*	

$$M\mathcal{H}$$

$$\mathcal{H} = \prod_{i=1}^n [h_i^{min}, h_i^{max}]$$

$$h_i$$

$$\mathcal{A}$$

$$\forall (x,y) \in \mathcal{D} : (\mathcal{A}(x)) = (x) = y$$

$$M\theta M'$$

$$|\theta'| < |\theta| \wedge (M') \geq (M) - \epsilon$$

$$\theta'_i = \begin{cases} 0 & |\theta_i| < \end{cases}$$

$$\theta'_{quantized} = \left(\frac{\theta}{\Delta} \right) \cdot \Delta$$

$$\mathcal{L}_{distill} = \alpha \mathcal{L}_{CE}(y, \hat{y}) + (1 - \alpha) \mathcal{L}_{KL}(p_T, p_S)$$

$p_T p_S$

$$W \approx UV^TW \in \mathbb{R}^{m \times n}, U \in \mathbb{R}^{m \times r}, V \in \mathbb{R}^{n \times r}$$

$$r \ll (m, n)$$

$M_0, clients, rounds$	
$M_{global} \leftarrow M_0$	▷
$t = 1rounds$	
$S_t \leftarrow (clients, frac = 0.1)$	▷
$k \in S_t$	
$M_k \leftarrow (M_{global}, data_k, epochs = 5)$	
$M_{global} \leftarrow \frac{1}{ S_t } \sum_{k \in S_t} M_k$	▷
M_{global}	

$C\sigma$

$$\epsilon = \frac{C\sqrt{2(1.25/\delta)}}{\sigma}$$

(ϵ, δ)

$$x\hat{y}A(x,\hat{y})$$

$$A_i=\frac{\partial \hat{y}}{\partial x_i}$$

$$|A_i|$$

$$\mathcal{L}_{EWC} = \mathcal{L}_{new} + \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_{A,i}^*)^2$$

$$F_i\theta_A^*$$

$$\theta$$

$$\theta^* = \sum_{\tau_i \sim p(\tau)}_{\theta} \mathcal{L}_{\tau_i}(U_{\tau_i}(\theta))$$

$$U_{\tau_i}\tau_i$$

$$tasks, innerlr, outerlr, iterations$$

$$\theta \leftarrow ()$$

$$i = 1 iterations$$

$$\mathcal{D}_{meta} \leftarrow (tasks)$$

$$\mathcal{T} \in \mathcal{D}_{meta}$$

$$\mathcal{D}_{train}, \mathcal{D}_{test} \leftarrow (\mathcal{T})$$

$$\theta' \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{T}}(\theta, \mathcal{D}_{train})$$

$$\mathcal{L}_{meta} \leftarrow \mathcal{L}_{\mathcal{T}}(\theta', \mathcal{D}_{test})$$

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \sum \mathcal{L}_{meta}$$

$$\theta$$

$$OD=\mu_{DSL}(O)$$

$$(D)\leftrightarrow (O)$$

$$D_1,D_2,\ldots,D_nD=D_1\times D_2\times\cdots\times D_n$$

$$(D)=\bigcup_{i=1}^n(D_i)$$

$$(D)=\bigotimes_{i=1}^n(D_i)$$

$$\mathcal{A}=(S,A,T,\delta)$$

$$S$$

$$A$$

$$T:S\times A\rightarrow S$$

$$\delta:S\rightarrow 2^A$$

$$3f+1f$$

agents, goal
proposals $\leftarrow \emptyset$
agent $\in$ *agents*
proposal $\leftarrow$ *agent.propose(goal)*
proposals.add(proposal)

$|proposals| > 1$
pairs $\leftarrow (proposals)$
 $(p_1, p_2) \in pairs$
 $p_1.rank > p_2.rank$
proposals.remove(p₂)

proposals.remove(p₁)

proposals.first()

▷

agents, command
primary $\leftarrow agents[leader]$
primary.broadcast(pre – prepare, command)
replica $\in agents[followers]$
replica.verify(command)
replica.broadcast(prepare, command)

$count(prepare) \geq 2f$
replica.broadcast(commit, command)

$count(commit) \geq 2f + 1$
replica.execute(command)
replica.broadcast(reply, result)

system

health $\leftarrow$ *system.checkhealth()*
health.hasfaults()
faults $\leftarrow$ *health.diagnose()*
fault $\in$ *faults*
repair $\leftarrow$ *system.planrepair(fault)*
repair.issafe()
system.executerepair(repair)

system.requesthumanintervention(fault)

(*monitorinterval*)

ND

$$v_i^{t+1} = wv_i^t + c_1r_1(p_{i,best} - x_i^t) + c_2r_2(g_{best} - x_i^t)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1}$$

vp_{i,best}g_{best}

$(\mathcal{S}, \mathcal{A}, P, R, \gamma)$

$\mathcal{S}$

$\mathcal{A}$

$P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$

$R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$

$\gamma \in [0, 1]$

objective, particles, iterations

swarm $\leftarrow$ (*particles*)

particle $\in$ *swarm*

particle.pbest $\leftarrow$ *particle.position*

particle.fitness $\leftarrow$ *objective*(*particle.position*)

gbest $\leftarrow$ $\min_p$ *p.fitness*

t = 1 *iterations*

particle $\in$ *swarm*

particle.velocity $\leftarrow$ *w* $\cdot$ *particle.velocity*

particle.velocity $\leftarrow$ *particle.velocity* + $c_1 r_1$ (*particle.pbest* - *particle.position*)

particle.velocity $\leftarrow$ *particle.velocity* + $c_2 r_2$ (*gbest* - *particle.position*)

particle.position $\leftarrow$ *particle.position* + *particle.velocity*

fitness $\leftarrow$ *objective*(*particle.position*)

fitness < *particle.fitness*

particle.fitness $\leftarrow$ *fitness*

particle.pbest $\leftarrow$ *particle.position*

fitness < *gbest.fitness*

gbest $\leftarrow$ *particle.position*

gbest

$$T = (O_{prev}, O_{new}, sig, ts)$$

O_{prev}

O_{new}

sig

ts

R

$$R = (H_O, H_A, , ,)$$

 $\overline{O}$
$$json \leftarrow (O)$$
$$cid \leftarrow (json)$$
$$tx \leftarrow (cid)$$

cid

cid

$$json \leftarrow (cid)$$
$$verified \leftarrow (cid)$$

verified

$$O \leftarrow (json)$$
 O $Err(Unverified)$

$O_{runtime}$	
$cachekey \leftarrow (O_{runtime})$	
$cache.contains(cachekey)$	
$cache.get(cachekey)$	▷
$A \leftarrow \mu(O_{runtime})$	▷
$compiled \leftarrow (A)$	▷
$optimized \leftarrow (compiled)$	▷
$cache.insert(cachekey, optimized)$	
$optimized$	

$O_{runtime}O_{static}$

$O_{runtime} \supseteq O_{static}$

K

```

input, strategies
profile  $\leftarrow$  (input)
selected  $\leftarrow$  (profile, strategies)
result  $\leftarrow$  selected.execute(input)
(selected, result)
result

profile, strategies
scores  $\leftarrow \emptyset$ 
strategy  $\in$  strategies
history  $\leftarrow$  (strategy, profile)
expected  $\leftarrow$  (strategy, profile, history)
scores.add(strategy, expected)

strategy scores[strategy]

```

$$A = \mu(O)$$

$$A = \mu(O)$$

$$A = \mu(O)$$

Codebase

AST $\leftarrow$ (*Codebase*)

entities $\leftarrow$ (*AST*)

▷

relations $\leftarrow$ (*AST*)

▷

constraints $\leftarrow$ (*AST*)

▷

O $\leftarrow$ (*entities, relations, constraints*)

closure $\leftarrow$ (*O*)

closure $\neq$ *Complete*

missing $\leftarrow$ (*O*)

O $\leftarrow$ (*missing*)

▷

O

$Ce \in \mathbb{R}^d$

$e = (C)$

$I_1, I_2, \dots, I_n$

$(I_1) = (I_2) = \dots = (I_n)$

$\mu((I_1)) = \mu((I_2)) = \dots = \mu((I_n))$

Codebase

embeddings $\leftarrow \emptyset$

file $\in$ *Codebase*

AST $\leftarrow$ (*file*)

node $\in$ *AST*

e $\leftarrow$ (*node*)

embeddings[*node*] $\leftarrow e$

clusters $\leftarrow$ (*embeddings*)

▷

entities $\leftarrow$ (*clusters*)

relations $\leftarrow$ (*embeddings*)

O $\leftarrow$ (*entities*, *relations*)

O

text

entities $\leftarrow$ (*text*)

▷

relations $\leftarrow$ (*text*)

▷

constraints $\leftarrow$ (*text*)

▷

triples $\leftarrow$ (*entities*, *relations*, *constraints*)

O $\leftarrow$ (*triples*)

▷

O

```

image
objects ← (image)                                ▷
layout ← (objects)                                ▷
text ← (image)                                     ▷
schema ← (objects, text)
O ← (schema)
O

```

$$E = \{(input_1, output_1), \dots, (input_n, output_n)\} O$$

$$\forall (i, o) \in E : \mu(O)(i) = o$$

O

```

examples
patterns ← ∅
(input, output) ∈ examples
diff ← (input, output)                                ▷
pattern ← (diff)                                       ▷
patterns.add(pattern)

O ← (patterns)
O

```

$O_{current}, O_{proposed}, proposer$
 $proposal \leftarrow (O_{current}, O_{proposed})$
 $proposal.id \leftarrow ()$
 $proposal.proposer \leftarrow proposer$
 $proposal.votes \leftarrow \emptyset$
 $proposal.deadline \leftarrow () + 7days$
 $proposal$

$proposal, voter, decision$
 $(voter) \wedge (proposal)$
 $proposal.votes.add((voter, decision))$
 $(proposal.votes)$
 $(proposal)$

P_1, P_2

$\exists t \in P_1.triples, t' \in P_2.triples : (t, t')$

$NFT = (O_{content}, owner, license, price)$

$O_{content}license$

$$v = M.m.p$$

$$M$$

$$m$$

$$p$$

$$O_11.0.0O_21.1.0$$

$$\mu(O_1) \subseteq \mu(O_2)$$

$$O_1O_2$$

$$\begin{array}{l} O_{old}, O_{new} \\ \Delta \leftarrow (O_{old}, O_{new}) \\ breaking \leftarrow \{d \in \Delta \mid d.type = Breaking\} \\ migration \leftarrow (breaking) \\ code \in \mu(O_{old}) \\ code' \leftarrow (code, migration) \\ (code', O_{new}) \end{array}$$

$$migration$$

$$f(O)$$

$$f(O) = w_1 \cdot (O) + w_2 \cdot (O) + w_3 \cdot \frac{1}{(O)}$$

```

population, generations
gen = 1generations
fitness ← (population)
selected ← (population, fitness)
of fspring ← (selected)
mutated ← (of fspring)
population ← (population ∪ mutated)

population.best()

```

$$\Delta O_{old}O_{new}$$

$$(O_{old}) \overset{reload}{\longrightarrow} (O_{new})$$

$$O_{new}O_{old}$$

$$O_{new}$$

$$O_{old}$$

$$O_1,O_2O=O_1\cup O_2$$

$$O=\{t\mid t\in O_1\vee t\in O_2\}$$

$O_{current}, O_{proposed}$	
$\Delta \leftarrow (O_{current}, O_{proposed})$	
(Δ)	
$O_{shadow} \leftarrow (O_{current}, \Delta)$	
$A_{shadow} \leftarrow \mu(O_{shadow})$	$\triangleright$
(A_{shadow})	
$comparison \leftarrow (A_{current}, A_{shadow})$	
$comparison.difference < threshold$	
(A_{shadow})	
$()$	
$Err()$	

$$O = O_1 \cap O_2 \qquad O = \{t \mid t \in O_1 \wedge t \in O_2\}$$

$$O = O_1 \setminus O_2 O_2 O_1 \qquad O = \{t \mid t \in O_1 \wedge t \notin O_2\}$$

$$O_1, O_2 \qquad \mu(O_1 \cup O_2) = \mu(O_1) \cup \mu(O_2)$$

$$O_1 \models \Sigma_1 O_2 \models \Sigma_2 \qquad O_1 \cup O_2 \models \Sigma_1 \cup \Sigma_2$$

$$A = (P, A)$$

$$P$$

$$A$$

OM

$M \models O$

MO

O

$clauses \leftarrow (O)$

$\triangleright$

$solver \leftarrow ()$

$clause \in clauses$

$solver.add(clause)$

$result \leftarrow solver.solve()$

$result = SAT$

$Ok(model)$

$Err(Unsatisfiable)$

O

$\nexists t_1, t_2 \in O : t_1 = (s, p, o) \wedge t_2 = (s, p, o') \wedge o \neq o'$

$(subject, predicate)$

O, Φ	
$M \leftarrow (O)$	$\triangleright$
$\phi \in \Phi$	
$\phi.type = LTL$	$\triangleright$
$result \leftarrow (M, \phi)$	
$\phi.type = CTL$	$\triangleright$
$result \leftarrow (M, \phi)$	
$result = Violated$	
$counterexample \leftarrow (M, \phi)$	
$Err(\phi, counterexample)$	
Ok	

P

$$\mu_P(O) = \sum_{A \in \{A' | A' = \mu(O)\}} (A, P)$$

AP

$$P = \{(a, o) \mid a \in A, o \in O, ao\}$$

O

$A \leftarrow \mu(O)$

$feedback \leftarrow (A)$

$feedback = Accept$

A

$feedback = Reject$

$concerns \leftarrow (feedback)$

$O' \leftarrow (O, concerns)$

$A \leftarrow \mu(O')$

$\Pi = (Template, Constraints)$

Template

Constraints

$\Pi, D_{source}, D_{target}$	
$instances \leftarrow (\Pi, D_{source})$	▷
$adapted \leftarrow (\Pi, D_{target})$	▷
$validated \leftarrow (adapted, D_{target})$	
$validated$	
$adapted$	
$Err(PatternNotApplicable)$	

$$\mathcal{M} \qquad \mathcal{M} : \mathcal{D} \rightarrow \mathcal{O}$$

$\{(D_i, O_i)\}$	
$\mathcal{M} \leftarrow ()$	
$= 1N$	
$(D, O) \in \{(D_i, O_i)\}$	
$\hat{O} \leftarrow \mathcal{M}(D)$	▷
$\mathcal{L} \leftarrow (O, \hat{O})$	▷
$\mathcal{M} \leftarrow \mathcal{M} - \alpha \nabla \mathcal{L}$	▷

$$\mathcal{M}$$

$$(O) = \alpha \cdot |O| + \beta \cdot H(O) + \gamma \cdot D(O)$$

$$|O|$$

$$H(O)$$

$$D(O)$$

$$\alpha, \beta, \gamma$$

$$({\cal O})=\frac{({\cal O})}{({\cal O})}$$

$$({\cal O})=\frac{({\cal O})}{({\cal O})}$$

$$O_1,O_2,\ldots,O_nS_1,S_2,\ldots,S_n$$

$$P(O)=\sigma\left(\sum_iw_i\cdot M_i(O)\right)$$

$$M_i(O)\sigma$$

$$({\cal P})=\frac{|\{O\mid P(O)=S(O)\}|}{n}>0.9$$

$$I_1, I_2, \dots, I_n$$

$$(I_1) = (I_2) = \dots = (I_n)$$

$$\mu((I_1)) = \mu((I_2)) = \dots = \mu((I_n))$$

<i>text</i>	
<i>entities</i> $\leftarrow$ (<i>text</i>)	▷
<i>relations</i> $\leftarrow$ (<i>text</i>)	▷
<i>constraints</i> $\leftarrow$ (<i>text</i>)	▷
<i>triples</i> $\leftarrow$ (<i>entities</i> , <i>relations</i> , <i>constraints</i>)	
<i>O</i> $\leftarrow$ (<i>triples</i>)	▷
<i>O</i>	

$$DER$$

$$O = \{(e_1, r, e_2) \mid e_1, e_2 \in E, r \in R\}$$

```

Ocurrent, Oproposed, proposer
proposal ← (Ocurrent, Oproposed)
proposal.id ← ()
proposal.proposer ← proposer
proposal.votes ← ∅
proposal.deadline ← () + 7days
proposal

proposal, voter, decision
(voter) ∧ (proposal)
proposal.votes.add((voter, decision))
(proposal.votes)
(proposal)

```

$$P_1, P_2 \qquad \exists t \in P_1.triples, t' \in P_2.triples : (t, t')$$

$$v = M.m.p$$

$$M$$

$$m$$

$$p$$

$$O_11.0.0O_21.1.0$$

$$\mu(O_1) \subseteq \mu(O_2)$$

$$O_1O_2$$

O_{old}, O_{new}
 $\Delta \leftarrow (O_{old}, O_{new})$
 $breaking \leftarrow \{d \in \Delta \mid d.type = Breaking\}$
 $migration \leftarrow (breaking)$
 $code \in \mu(O_{old})$
 $code' \leftarrow (code, migration)$
 $(code', O_{new})$

 $migration$

$\Delta O_{old} O_{new}$

$(O_{old}) \xrightarrow{reload} (O_{new})$

$O_{new} O_{old}$

O_{old}

$$O_1, O_2 O = O_1 \cup O_2$$

$$O = \{t \mid t \in O_1 \vee t \in O_2\}$$

$$O = O_1 \cap O_2$$

$$O = \{t \mid t \in O_1 \wedge t \in O_2\}$$

$$O = O_1 \setminus O_2 O_2 O_1$$

$$O = \{t \mid t \in O_1 \wedge t \notin O_2\}$$

$$O_1, O_2$$

$$\mu(O_1 \cup O_2) = \mu(O_1) \cup \mu(O_2)$$

$$O_1 \models \Sigma_1 O_2 \models \Sigma_2$$

$$O_1 \cup O_2 \models \Sigma_1 \cup \Sigma_2$$

$$OM$$

$$M \models O$$

$$MO$$

$$O$$

$$\nexists t_1, t_2 \in O : t_1 = (s, p, o) \wedge t_2 = (s, p, o') \wedge o \neq o'$$

O

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$\triangleright$

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$(subject, predicate)$

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$$\mu_P(O) = \sum_{A \in \{A' | A' = \mu(O)\}} (A, P)$$

O

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A

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$O' \leftarrow (O, concerns)$

$A \leftarrow \mu(O')$

$\Pi = (Template, Constraints)$

$Template$

$Constraints$

$\Pi, D_{source}, D_{target}$

$instances \leftarrow (\Pi, D_{source})$

▷

$adapted \leftarrow (\Pi, D_{target})$

▷

$validated \leftarrow (adapted, D_{target})$

$validated$

$adapted$

$Err(PatternNotApplicable)$

$$(O) = \alpha \cdot |O| + \beta \cdot H(O) + \gamma \cdot D(O)$$

$$|O|$$

$$H(O)$$

$$D(O)$$

$$\alpha,\beta,\gamma$$

$$(O)=\frac{(O)}{(O)}$$

$$(O)=\frac{(O)}{(O)}$$

$$O_1, O_2, \ldots, O_n S_1, S_2, \ldots, S_n$$

$$P(O)=\sigma\left(\sum_iw_i\cdot M_i(O)\right)$$

$$M_i(O)\sigma$$

$$(P)=\frac{|\{O\mid P(O)=S(O)\}|}{n}>0.9$$

$$O_{fork} = O_{upstream} \cup \Delta_{local} O'_{upstream} = O_{upstream} \cup \Delta_{upstream}$$

$$O_{merged} = O'_{upstream} \cup \Delta_{local}$$

$$A$$

$$OB$$

$$A=\mu(O)A\models B$$

$$A(t)\in Bt$$

$$A=\mu(O)$$

$$O$$

$$\mu$$

$$A$$

$\mu_1\mu_5$

$$O\mu(O)A$$

$$\mu=\mu_5\circ\mu_4\circ\mu_3\circ\mu_2\circ\mu_1$$

$$\mu_i$$

$$y_i=\mu_i(x_i)$$

$$A_1=\mu_1(O)$$

$$A_2=\mu_2(A_1)$$

$$A_3=\mu_3(A_2)$$

$$A_4=\mu_4(A_3)$$

$$A=\mu_5(A_4)=\mu(O)$$

$$\mu_i\mu$$

$$OTE A = \mu(O)TE$$

$$\mu_1O(E)=TT$$

$$\mu_2ET$$

$$\mu_3$$

$$\mathcal{H}^{\mathcal{A}}_{\mathcal{B}}(\mathcal{C})=\mathcal{H}^{\mathcal{A}}_{\mathcal{B}}(\mathcal{C})\cap\mathcal{H}^{\mathcal{A}}_{\mathcal{B}}(\mathcal{C})$$

$$n=3f+1$$

$$vv'v\neq v'2f+1f+1vv'$$

$$f2f+12f+1$$

$$\Phi(O,A)\geq 0.98$$

$$\Phi(O,A)=\frac{MI(O,A)}{H(O)}$$

$$MI(O,A)OA$$

$$A=\mu(O)\mu$$

$$H(A|O)=0$$

$$MI(O,A)=H(A)-H(A|O)=H(A)$$

$$\Phi(O,A)=\frac{H(A)}{H(O)}\geq 0.98$$
